

Interacting Multipoles of Rank 1 and 2:

$$T_{\alpha\beta\gamma}\mu_{\alpha}^A\Theta_{\beta\gamma}^B = R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot \mu_{\alpha}^A \cdot \Theta_{\beta\gamma}^B \\ +3 \cdot R_z^2 \cdot R_{\alpha} \cdot \delta(\beta, \gamma) \cdot \mu_{\alpha}^A \cdot \Theta_{\beta\gamma}^B \\ +3 \cdot R_z^2 \cdot R_{\beta} \cdot \delta(\alpha, \gamma) \cdot \mu_{\alpha}^A \cdot \Theta_{\beta\gamma}^B \\ +3 \cdot R_z^2 \cdot R_{\gamma} \cdot \delta(\alpha, \beta) \cdot \mu_{\alpha}^A \cdot \Theta_{\beta\gamma}^B \end{bmatrix} \quad (1)$$

Simplify deltas:

$$T_{\alpha\beta\gamma}\mu_{\alpha}^A\Theta_{\beta\gamma}^B = R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot \mu_{\alpha}^A \cdot \Theta_{\beta\gamma}^B \\ +3 \cdot R_z^2 \cdot R_{\alpha} \cdot \delta(\beta, \beta) \cdot \mu_{\alpha}^A \cdot \Theta_{\beta\beta}^B \\ +3 \cdot R_z^2 \cdot R_{\beta} \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^A \cdot \Theta_{\alpha\beta}^B \\ +3 \cdot R_z^2 \cdot R_{\gamma} \cdot \delta(\alpha, \alpha) \cdot \mu_{\alpha}^A \cdot \Theta_{\alpha\gamma}^B \end{bmatrix} \quad (2a)$$

$$\xRightarrow{\text{cleaned up}} R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot \mu_{\alpha}^A \cdot \Theta_{\beta\gamma}^B \\ +3 \cdot R_z^2 \cdot R_{\alpha} \cdot \mu_{\alpha}^A \cdot \Theta_{\beta\beta}^B \\ +3 \cdot R_z^2 \cdot R_{\beta} \cdot \mu_{\alpha}^A \cdot \Theta_{\alpha\beta}^B \\ +3 \cdot R_z^2 \cdot R_{\gamma} \cdot \mu_{\alpha}^A \cdot \Theta_{\alpha\gamma}^B \end{bmatrix} \quad (2b)$$

$$\xRightarrow{\text{reduced indices}} R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot \mu_{\alpha}^A \cdot \Theta_{\beta\gamma}^B \\ +3 \cdot R_z^2 \cdot R_{\alpha} \cdot \mu_{\alpha}^A \cdot \Theta_{\beta\beta}^B \\ +3 \cdot R_z^2 \cdot R_{\beta} \cdot \mu_{\alpha}^A \cdot \Theta_{\alpha\beta}^B \\ +3 \cdot R_z^2 \cdot R_{\beta} \cdot \mu_{\alpha}^A \cdot \Theta_{\alpha\beta}^B \end{bmatrix} \quad (2c)$$

$$\xRightarrow{\text{added up}} R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_{\alpha} \cdot R_{\beta} \cdot R_{\gamma} \cdot \mu_{\alpha}^A \cdot \Theta_{\beta\gamma}^B \\ +3 \cdot R_z^2 \cdot R_{\alpha} \cdot \mu_{\alpha}^A \cdot \Theta_{\beta\beta}^B \\ +6 \cdot R_z^2 \cdot R_{\beta} \cdot \mu_{\alpha}^A \cdot \Theta_{\alpha\beta}^B \end{bmatrix} \quad (2d)$$

Simplify R:

$$T_{\alpha\beta\gamma}\mu_{\alpha}^A\Theta_{\beta\gamma}^B = R_z^{-7} \cdot -\frac{1}{3} \cdot \begin{bmatrix} -15 \cdot R_z \cdot R_z \cdot R_z \cdot \mu_z^A \cdot \Theta_{zz}^B \\ +3 \cdot R_z^2 \cdot R_z \cdot \mu_z^A \cdot \Theta_{\beta\beta}^B \\ +6 \cdot R_z^2 \cdot R_z \cdot \mu_{\alpha}^A \cdot \Theta_{z\alpha}^B \end{bmatrix} \quad (3a)$$

$$\xRightarrow{\text{cleaned up}} R_z^{-7} \cdot -\frac{1}{3} \cdot [+6 \cdot R_z^3 \cdot \mu_{\alpha}^A \cdot \Theta_{z\alpha}^B] \quad (3b)$$

Simplify Dipoles:

$$T_{\alpha\beta\gamma}\mu_{\alpha}^A\Theta_{\beta\gamma}^B = R_z^{-7} \cdot -\frac{1}{3} \cdot [+6 \cdot R_z^3 \cdot \mu_y^A \cdot \Theta_{yz}^B] \quad (4a)$$

$$\xRightarrow{\text{cleaned up}} R_z^{-7} \cdot -\frac{1}{3} \cdot [+0] \quad (4b)$$

Combining everything:

$$T_{\alpha\beta\gamma}\mu_{\alpha}^A\Theta_{\beta\gamma}^B = +0 \quad (5)$$